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Problem Set 2

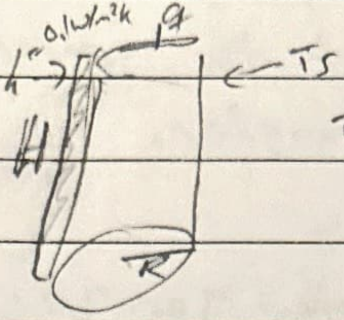
Date: 2/10/26

Acknowledgement: No help

1) Known: $R = 0.5m$ $H = 1m$ $T_w = 50°C = T_s$ $T_\infty = 20°C$ $h = 15 W/m^2k$

Find: a) q needed b) q needed w/ 2cm insulation c) compare a and b

Schematic:



T_∞, h

Assumptions: radiation is negligable
Steady state

$$A = \pi(0.5)^2 + \pi(2(0.5)(1)) = 3.93m$$

Analysis: a) $q_{conv} = hA(T_s - T_\infty) = 15(3.93)(30-20) = \boxed{1.77kW}$

b) $q_{cond} = -kA\frac{dT}{dx} = q_{conv} = hA(T_s - T_\infty)$

$-0.1A\frac{(T_{in} - T_{out})}{0.02m} = 15A(T_{out} - T_\infty)$ $0.1(50 - T_{out})/0.02 = 15(T_o - 20)$

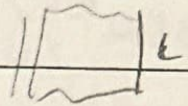
$T_o = 27.5°C$ $q = 15(A)(27.5-20) = \boxed{441.79 W}$

c) with insulation takes 75% less power

2) Known : $L = 10\,cm$ $K = 1\,W/m \cdot k$ $T_\infty = 20°c$ $h = 100\,W/m^2k$ $q'' = 1000\,w/m^2$

find} a) T_{in}, T_{out} b) max q'' for $T_{(L)} < 200°c$ c) 3 stmnts to use higher q'' with low T

Schematic :



assumptions : Inf. thin, no radiation, steady state

Analysis: a) $q''_{cond} = -k \frac{dT}{dx}$ $q''_{conv} = h(T_s - T_\infty)$

$1000 = 100(T(L) - 20°c)$ $\boxed{T(L) = 30°c}$

$1000 = -1 \frac{30°c - T(0)}{0.1m}$ $\boxed{T(0) = 130°c}$

b) $T(L) = \frac{q''}{h} + T_\infty$ $T(0) = \frac{q''L}{k} + T(L)$

$T(0) = 200°c = T_\infty + \frac{q''}{h} + \frac{q''L}{k} = 20 + \frac{q''}{100} + \frac{q''0.1}{1}$

$\boxed{q'' = 1636.4\,W/m^2}$

c) To increase q'' without increasing T could be to increase h, increase k, decrease L or lower T_∞.

Date:

3) known: focused laser isotherm provided

find: a) draw heat flux vectors b) left or right is larger

schematic/analysis: assumptions steady state, uniform material properties.

a)

perpindicular to isotherms

b) left is larger heat flux because the isotherms are spaced closer together.